Response to the review of 2026-10-19 (commit ‘bfc797ef43‘)

Draft of 2026-10-19

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Reviewed commit: `bfc797ef43c28d6346ce68ce51846e20fb2b2e77`. Revision: the aligned-policy round of 2026-10-20. Every number below is regenerated by the scripts named in the last column; the round’s artifacts are under `results/wrr_submission_ready/major_revision_20261020/`.

The review’s central judgement was that the same method’s risk control, actual replay and environmental judgement had not been closed, because the admitted object was the policy reconstruction while the calibrated event was the certificate error. We agree, and we rebuilt the chain rather than patching the prose. The rebuild changed the headline: with the right object and a deployment-consistent tie rule, the fitted score’s advantage over the peak-window rule is 7.0–8.3 points on the external panel, not 2.6–3.3, and it clears the pre-specified five-point margin. It still does not beat the certificate bound, and on real calendars no screen is needed. The paper is now written as a method paper with a named regime boundary, not as a claim of general superiority.

id	severity	status	what changed
F01	Blocker	closed	every stage now reads `abserr_policy`; the certificate error survives only as a ranking score

id	severity	status	what changed
F02	Major	closed	population risk and finite-batch error are separated; the 8.5 % figure is retracted as a misreading
F03	Major	closed	no jitter; whole tie blocks admitted in calibration, evaluation and deployment; the fixed-70 % rule uses a stable case key
F04	Major	closed	`reconstruct_all` is a first-class arm; the regime boundary is measured, not asserted
F05	Major	closed	all three fold seeds; confusion matrix, refusal cost, paired intervals, station-year weighting
F06	Major	closed	the pre-specified margin is now met against the strong transparent baseline; the route decision is stated
F07	Major	closed	manuscript and SI rewritten around one chain; old certificate numbers are historical
F08	Major	closed	the “position governs” law is replaced by a conditional statement with the reconstructor counterexample
F09	Major	accepted as scope	the external comparison is a cross-fit on an opened panel; no sealed claim is made
F10	Minor	closed	counts, units and share-versus-points fixed; the measurement-stratum gap is disclosed
F11	Minor	partly closed	scripts, fingerprints and manifests are in place; the author block remains a human gate

F01 — final reconstruction object and failure event

The two failure events disagree on 898 of 13,017 external cases (6.90 %): 408 cases fail only under the policy reconstruction and 490 only under the certificate. On the

development panel they disagree on 830 of 40,704 cases (2.04 %).

We rebuilt the full protocol on the policy error. `scripts/330` asserts before any computation that `abserr_policy = |hottest30_<policy_choice> − truth_hottest30|` on every finite row, so an alignment error aborts the run. The threshold, the evaluated event and the reported error are now the same column at every stage; the certificate bounds are used only as ranking scores, which the review explicitly allowed.

Consequence, measured on the same scores and folds (`scripts/334`):

panel	event / tie	fitted − peak (pp)
external	certificate / jitter (old design)	+2.8
external	certificate / whole block	+2.7
external	policy / whole block (new)	+7.2
development	certificate / jitter (old design)	+8.6
development	policy / whole block (new)	+9.7

About 4.5 points of the external change come from the object and about 1.3 from the tie rule. The fitted score now keeps 74.0 % of the external panel at the 5 % cap (peaks 73.6–74.5 across seeds) against 66.7 % for the peak-window rule, and the gain clears the pre-specified five-point margin at all three fold seeds. It does **not** beat the certificate bound: +1.6 points [0.2, 3.1] on the external panel, −2.6, −1.7 and −2.5 points on development.

Evidence: `p0_01_aligned_policy_admission/`, `p0_05_repair_attribution/`.

F02 — what the selector’s calibration means

The 8.5 % reported previously was the probability that a finite fresh test batch exceeds the cap, not the probability that the selected threshold’s population risk does. We re-ran 2,000 repetitions per scenario with the deployed rule imported from the analysis code, and measured the population risk of the selected threshold on an independent 2,000,000-case reference draw.

scenario	P(true risk > 5 %)	P(finite batch > 5 %)	retention
baseline	0.004	0.078	0.185
near cap	0.006	0.075	0.288
unequal groups	0.004	0.046	0.296
twelve groups	0.024	0.122	0.153

The rule is conservative, not miscalibrated. The price is retention: a pointwise bound keeps 35.9 % in the baseline scenario but violates the cap about six times as often (7.4 %). We also checked the extreme percentile: re-running the selection with 10,000 instead of 1,000 bootstrap draws changes the chosen threshold in 42 of 150 fold-by-score cells, moving the fitted score’s retention by at most 1.0 point and the discrete rules by at most 3.9 points. The paper now describes the threshold as a calibration decision at this granularity, not a stable optimum.

Evidence: `p0_02_selection_calibration/`, `p0_01_.../threshold_stability.csv`.

F03 — one tie rule from calibration to deployment

No jitter is added anywhere. A threshold is an observed value of the raw score, and admission is $s \le \tau$, so a tie block is admitted whole in calibration, in evaluation and at deployment. The fixed-70 % comparison, which is a second deployment rule, breaks ties with a stable case key (`network|station|year|realisation`) rather than row order. The ablation above bounds the cost of the old inconsistency at about 1.3 points on the external panel.

F04 — does screening pay against “no screen”?

Yes on designed outages, no on the real calendars. On the external panel, `reconstruct_all` fails at 15.3 %, so the 5 % cap binds and retention is the comparison. On the 418 real calendars the median gap is 7 days, the median missing fraction is 2.2 %, and 5.7 % of candidates hide more than half the peak window; admitting every reconstruction puts 2.6–4.1 % of records over the tolerance, already under the cap. The screens keep 93–96 % and reduce the failure rate to 0.5–1.7 %; the fitted score’s advantage over the peak-window rule falls to 0.2–2.4 points, and only two of nine level-by-seed intervals exclude zero. We did not tighten the cap to make screening look necessary; the regime boundary is the result.

Evidence: `p0_03_real_calendar_adoption/`, manuscript §4.4 and §5.3.

F05 — the environmental judgement

The hot-year analysis now covers all three fold seeds, both hot thresholds and both weightings, with a full confusion table and paired group-clustered intervals. At 20 °C on the external panel, the fitted score’s total deviation of the admitted hot-year share is +0.7 points [−0.3, +1.6]; the unique-station-year weighting gives +0.9 [0.0, 1.8] and leaves the ordering unchanged. The cost that a signed bias hides is the refusal column: the fitted score leaves out 24.5 % of true hot years, the certificate bound 26.2 %, the peak-window rule 33.2 % and the gap-length rule 59.5 %, against 0 % for reconstructing everything (which instead misjudges 1.3 % of hot years as cool). The paper reports both columns.

Evidence: `p0_04_hot_year_closure/`, SI Table S7.

F06 — innovation and the practical threshold

The pre-specified margin was five points against the peak-window rule, not against the length rule. With the object repaired the margin is met: +8.0, +7.3 and +7.0 points on the external panel, +10.3, +10.1 and +9.7 on development. It is not met against the certificate bound, which is a strong transparent alternative and keeps more records on development. The paper therefore claims a gain over a baseline that ignores support, not over every alternative, and it names the regime in which even that gain has no decision value. This follows the pre-declared rule rather than a post-hoc redefinition: the margin was not lowered, and the arm that beats us is reported.

F07 — one chain in the text

The manuscript was rewritten rather than amended. Methods describe the aligned protocol; Results follow the four reasoning steps (phenomenon, reconstructor, mechanism, decision); the old certificate-screen numbers appear only in the SI as history. The SI was rewritten to match, with new tables for the repair attribution, the selector calibration, retention and the hot-year composition.

F08 — mechanism and causal boundary

The claim “position, not length, governs” is withdrawn. What the crossed design supports is: the sign of the support effect is the same for every reconstructor, its size ranges from 0.24 to 0.64 °C at 30 days, and linear interpolation retains a length effect of 1.16 °C between 90 and 30 days even at full hidden support. The mechanism is window misalignment (0.28–0.70 °C when the reconstructed window lands away from the truth), reinforced by residual persistence (median lag-1 autocorrelation 0.84–0.88; higher-autocorrelation cases carry 0.16–0.36 °C more error under three methods). A competing peak does not order the error and is reported as a negative control.

F09 — validation scope

Unchanged and now stated in the abstract’s last sentence and in the limitations: the external panel is already opened, every comparison on it is a cross-fit within that panel, and no claim of a sealed confirmation is made. The real- calendar replay is a development-period probe with 418 groups in 14 networks.

F10 — units, eligibility and the SI

All counts now come from the aligned run: 5,114 complete station-years (146 networks, 457 stations); 40,704 development and 13,017 external designed cases (53,721 total); 2,422 incomplete real station-years; 418 replay groups. Shares are reported as fractions, differences in points. The measurement-stratum check that was open in the first draft of this response is now run: thresholds re-selected inside the documented-daily-mean subset give a fitted-vs-peak advantage of 8.0/7.3/7.3 points on the external subset and 8.8/8.7/10.4 on development, so the ranking does not depend on the project-aggregate or undocumented providers (`p1_measurement_subset/`).

F11 — FAIR and administration

Each analysis writes a protocol fingerprint (script hash, input hashes, git head), the figure sources are written next to the figures, and the build refuses to call the manuscript submission-ready while the author block is a placeholder. We did not independently re-derive every figure from the raw daily observations in this round, and we do not claim that we did.

What remains open

1. **A pristine panel.** None remains for this method. The paper is a cross-fit method study with a stated scope, and a sealed confirmation would need new data.
2. **The measurement-stratum check.** Re-estimating the aligned advantage inside the documented-daily-mean subset is a one-day analysis we flag rather than assume.
3. **The author block.** Human gate; names, affiliations, funding and ORCID identifiers are not inferred from a version-control alias.